

[X is the next letter after Q]

127. There are three drawers in a table. One contains two gold coins, another two silver coins and the third, a silver coin and a gold coin. One of the drawers is pulled out and a coin is taken out. It turns out to be a silver coin What is the probability of drawing a gold coin, if one of the other two drawers is pulled out next and one of the coins in it is drawn at random?

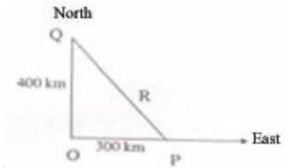
(a) 37.5% (b) 50%
(c) 62.5% (d) 75%

Ans. (a) : According to above instruction after a silver coin pulled out reamed two drawers in table is gold coin. For finding the silver coin only drawer 1 and 3 remains in consideration because the open drawer in any case cannot be the drawer that have only gold coins.

Now the probability of next coin being a gold coin = $\frac{1}{2}$

= 50%.

128. In the following figure.



P is 300 km Eastward of O and Q is 400 km North of O.R is exactly at the middle of Q and P. The distance between Q and R is:

(b) 250 km (c) 300 km
(d) 350 km (a) $250\times\sqrt{\text{km}}$

Ans. (a) : OP^2+OQ^2

$=\sqrt{(300)^2+(400)^2}$

$=\sqrt{100^2\left(3^2+4^2\right)}$

$=\sqrt{100^2\times5^2}$

$=100\times5=500$

R is being in the midway of PQ so QR = 250 km.

129. When three coins are tossed together the probability that all coins have the same face up is:

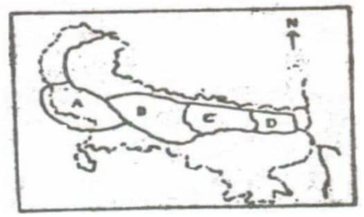
(a) $\frac{1}{3}$ (b) $\frac{1}{6}$
(c) $\frac{1}{8}$ (d) $\frac{1}{12}$

Ans. (c) : Probability of Head or Tail on the upper side for a coin = 1/2

∴ Probability of same side on the upper side for the

Three coins = $\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}=\left(\frac{1}{2}\right)^3=\frac{1}{8}$

130. Consider the map give below:



The predominant languages spoken in the areas marked A, B,C and D are respectively :

(a) Brajbhasha, Bhojpuri, Maithili and Awadhi
(b) Awadhi, Maithili, Bhojpuri and Brajbhasha
(c) Brajbhasha, Awadhi, Bhojpuri and Maithili
(d) Maithili, Bhojpuri, Brajbhasha and Awadhi

Ans. (c) : According to the given map the area marked 'A' is Mathura area of Uttar Pradesh, where Brajbhasha is spoken. 'B' is marked by Faizabad of U.P., and the language spoken is 'Awadhi'. 'C' area is 'Arah' area of Bihar where 'Bhojpuri' is spoken. 'D' refers to north Bihar where 'Maithili' is predominantly spoken

131. The number of students in two sections, A and B having different heights is shown in the following Table:

High (in meters)	Number of students with that height	
	in section A	in section B
1.55	3	2
1.60	7	6
1.62	12	14
1.65	15	14
1.68	8	9
1.71	6	5
1.75	3	4

The ration of the number of students of a particular height in Section A to that in Section B is the maximum for height of

(a) 1.55 m (b) 1.60 m
(c) 1.65 m (d) 1.71 m

Ans. (a) : Ratio of students of section A to B is maximum for 1.55 category height, it is 3/2 = 1.55

132. The world's fastest computer has been able to perform (as of Dec. 1996)

(a) 10^6 operations per second
(b) 10^9 operations per second
(c) 10^{12} operations per second
(d) 10^{15} operations per second

Ans. (c) : As of Dec. 1996 Hitachi Machine, the SR 2201 made by the Hitachi corporation of Japan was the fastest computer and was able to perform more than 220 billion mathematical operations a second, which is almost equal to 10^{12} operations per second.

133. The Basque separatist organisation is active in:

(a) Russia (b) Cyprus
(c) Portugal (d) Spain

Ans. (d) : The Basque conflict, also known as the Spain–ETA conflict, was an armed and political conflict from 1959 to 2011 between Spain and the Basque National Liberation Movement, a group of social and political Basque organizations which sought independence from Spain and France.

134. The following table shows the percent change in the amount of sales (in rupees) at different retail stores in a given neighbourhood market in the period 1993 to 1995:

Retail Store	Percent Change	
	1993 to 1994	1994 to 1995
Anshu	+10	
Borna	–20	+9
Calpo	+5	+12
Dilip	–7	–15
Elegant	+17	–8

If the sales at Anshu Store amounted to Rs. 8 Lakhs in 1993. Then the amount of sales (in lakhs of rupees) at that store in 1995 was:

(a) 7.92 (b) 8.00
(c) 8.80 (d) 9.68

Ans. (a) : If there are two changes of X% & Y%

Then net changes = $X+Y+\frac{XY}{100}$

Net Change = $10-10+\frac{10\times(-10)}{100}=-1\%$

Amount of sales in 1995 = $8\left(1-\frac{1}{100}\right)=7.92$

Hence option (a) will be correct.

135. If the numbers representing volume and surface area of a cube are equal, then the length of the edge of the cube in terms of the unit of measurement will be:

(a) 3 (b) 4
(c) 5 (d) 6

Ans. (d) : Surface area of cube = $6a^2$ [a = side of cube]

Volume of cube = a^3

So, as per question,

Surface Area of cube = volume of cube

$6a^2 = a^3$

So,

a = 6

136. The length, breadth and height of a room are'/' 'b' and 'h' respectively. The perimeter of the ceiling expressed as a percentage of the total area of the four walls is:

(a) 100 h (b) 100/h
(c) h (d) h/100

Ans. (b) : The area of room covered by 4 walls of length 'L' breadth 'b' and height 'h' is.

A = 2 [l + b] × h

Whereas perimeter of ceiling is 2 (l + b)

So, $\frac{\text{Perimeter of ceiling}}{\text{Area of the room}}\times100$

$=\frac{2(l+b)}{2(l+b)\times h}\times100$

$=\frac{100}{h}$

137. Seven persons P, Q, R, S, T U and V participate in and finish all the events of a series of swimming races. There are no ties at the finish of any of the events. V always finishes somewhere ahead of Q. Either R finishes first and T finishes last or S finishes first and U or Q finishes last. If in a particular race V finished fifth, then which one of the following would be true?

(a) S finishes firsts (b) R finishes second
(c) T finishes third (d) R finishes fourth

Ans. (a) : According to question the right sequence is

First ⇒ S
Second ⇒ U
Third ⇒ R
Fourth ⇒ T
Fifth ⇒ V
Sixth ⇒ P
Seventh ⇒ Q

So S finishes first hence correct option is (a).

138. Consider the following:

1. Saxena, David, Jain and Kumar were District Collectors at places P, Q, R and S respectively in 1970.
2. In 1972 they were transferred. Saxena and Jain interchanged places. Kumar and David also interchanged places.
3. One year later in 1973, they were again transferred such that David and Jain interchanged places and Saxena and Kumar were also interchanged.

What should be the next round of transfers so that all the four persons could have been posted at all the four places?

(a) Interchange Saxena and David as well as Jain and Kumar